

Section 4: Structural Unavoidability (C4)

Assumptions Used

Assumption 1 (Current Sectoral Penetration (A3)). *As of 2024, AI systems are operationally embedded in the following four critical societal systems, each belonging to a distinct sector:*

1. **Energy** (E): *AI manages load-balancing in national power grids (EU, US, China). Annual operating budget $B_E \approx \$10^{10}$ (US grid management).*
2. **Healthcare** (H): *FDA-cleared AI diagnostic tools operate in $>40\%$ of US hospitals; AlphaFold2 used by $>1M$ researchers. Annual operating budget $B_H \approx \$10^{11}$ (US hospital IT).*
3. **Fiscal** (F): *AI drives fraud detection at IRS and HMRC. Annual operating budget $B_F \approx \$10^9$ (tax-system IT).*
4. **Education** (D): *AI adaptive learning platforms serve $>50M$ K-12 students. Annual operating budget $B_D \approx \$10^9$ (EdTech at scale).*

Each system is operationally critical: sustained failure causes measurable societal harm (blackouts, diagnostic errors, tax fraud, learning disruption). For each, AI has been in active use for ≥ 3 years as of 2024, with integration depth $d_X(0) \geq 100$ (estimated number of dependent sub-processes per system).

Assumption 2 (Super-Linear Switching Costs (A4)). *Let $d \geq 0$ denote AI integration depth (number of dependent sub-processes). The switching-cost function $S : [0, \infty) \rightarrow [0, \infty)$ satisfies:*

1. Monotone: *S is strictly increasing; $S(0) = 0$.*
2. Super-linear: *There exists $c > 0$ such that $S(d) \geq cd^2$ for all $d \geq 1$.*
3. Depth grows with use: *For every period of active use, $d(t+1) \geq d(t) + 1$.*
4. Budget calibration: *For each system $X \in \{E, H, F, D\}$, the constant c and the current depth $d_X(0) \geq 100$ jointly satisfy $cd_X(0)^2 \geq B_X$, i.e., current switching cost already meets or exceeds one full year's operating budget. (Empirical basis: multi-year AI integration in critical systems requires dedicated teams, regulatory re-approval, and full pipeline redesign whose cost is at minimum comparable to annual IT spend.)*

Assumption 3 (No Viable Non-AI Substitute (A4b)). *For each system $X \in \{E, H, F, D\}$, no non-AI system can replicate the critical AI-performed functions at acceptable cost and quality within a 5-year horizon. (Basis: pre-AI methods were already replaced because they could not meet current operational demands at scale; the volume and complexity of tasks handled by AI in each system now exceed what pre-AI methods were designed for.)*

Statement to Prove (C4)

Theorem 1 (Structural Unavoidability). *Under Assumptions ??, ??, and ??:*

- (i) *AI is operationally embedded in at least four independent critical societal systems at integration depth $d_X(0) \geq 100$.*
- (ii) *For each such system, the current switching cost $S(d_X(0))$ is at least as large as the system's annual operating budget B_X , making removal financially prohibitive.*
- (iii) *Continued active use of any system increases its integration depth and therefore its switching cost, creating a self-reinforcing lock-in dynamic.*
- (iv) *Combining (ii), (iii), and ??: AI's role in these systems is structurally unavoidable — removal is both financially prohibitive and operationally infeasible, and the cost of removal only increases over time.*

Proof. Step 1: Claim (i) — four embedded critical systems.

By A3, AI is operationally embedded in systems E, H, F, D . These belong to four distinct societal sectors (energy, healthcare, fiscal, education) and are operationally critical by A3. Each has current integration depth $d_X(0) \geq 100$ per A3. The four systems are sector-distinct, establishing (i).

Step 2: Claim (ii) — financially prohibitive switching cost.

By A4.2, for any system X with $d_X(0) \geq 1$:

$$S(d_X(0)) \geq c d_X(0)^2.$$

By A4.4 (budget calibration), $c d_X(0)^2 \geq B_X$. Therefore:

$$S(d_X(0)) \geq B_X.$$

Switching costs equal or exceed the annual operating budget for every system $X \in \{E, H, F, D\}$. We define “financially prohibitive” as $S(d_X) \geq B_X$; this condition holds for all four systems. Claim (ii) is established.

Step 3: Claim (iii) — self-reinforcing feedback loop.

Fix any system X in continued active use. By A4.3, for each additional period of use: $d_X(t+1) \geq d_X(t) + 1$. By induction, for all $n \geq 1$:

$$d_X(t+n) \geq d_X(t) + n > d_X(t).$$

Since $d_X(t+n) > d_X(t) \geq 0$ and S is strictly increasing (A4.1):

$$S(d_X(t+n)) > S(d_X(t)).$$

Furthermore, by A4.2:

$$S(d_X(t+n)) \geq c(d_X(t) + n)^2,$$

which grows without bound as $n \rightarrow \infty$. This is the lock-in feedback loop: continued use $\Rightarrow$ deeper integration $\Rightarrow$ strictly higher switching cost. Claim (iii) is established.

Step 4: Claim (iv) — structural unavoidability.

We have established (ii): removal is financially prohibitive ($S(d_X) \geq B_X$) for every system X right now. By (iii), this cost will only increase as long as the systems remain active. By A4b, no non-AI substitute exists that could replace the AI’s critical functions at acceptable cost and quality within a 5-year horizon.

Jointly: (a) removing AI now costs $\geq B_X$ (prohibitive), (b) waiting only increases this cost (by claim iii), and (c) there is no viable alternative (A4b). Together these conditions mean there is no rational pathway to AI removal: immediate removal is financially infeasible, delayed removal is even more costly, and substitution is technically infeasible. Therefore AI’s role in these systems is structurally unavoidable and the infrastructure is AI-dependent in a structurally irreversible sense. Claim (iv) is established.

Conclusion. All four claims follow from A3, A4 (parts 1–4), and A4b. □